MIL-M-382B(MU)
22 July 1970
SUPERSEDING
JAN-M-382A
23 June 1949

MILITARY SPECIFICATION

MAGNESIUM POWDER
(for Use in Ammunition)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers powdered magnesium suitable for use in ammunition.

1.2 Classification. Magnesium powder shall be furnished in the following types, grades, and granulations:

1.2.1 Type.

Type I - Flaked and/or chip
Grade A (96 percent minimum magnesium content)
Grade B (86 percent minimum magnesium content)

Type II - Oblong chip with rounded edges
(96 percent minimum magnesium content)

Type III - Atomized
(98 percent minimum magnesium content)

FSC 6810

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1.2.2 Granulation.

Granulation No.	Nominal mesh size
1	40 to 80
2	40 to 80 (alternate)
3	50 to 100
4	50 to 100 (Army)
5	50 to 120
6	80 to 120
7	100
8	120 to 200
9	140
10	200
11	80 to 200
12	120 to 200 (Army)
13	20 to 50
14	50 to 100
15	100 to 200
16	200 to 325
17	50 to 100
18	30 to 50

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Standard for Testing Purposes

MILITARY

MIL-C-6054 - Container: Steel, Shipping

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

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(Copies of specifications, standards, drawings, and publications required by cuppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C., 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D1193-66 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I (grade A), types II and III. Type I (grade A), type II and III magnesium powder shall be manufactured from primary magnesium metal containing not less than 99.80 percent metallic magnesium. Mica fillers or other adulterants shall not be used and "bag house dust" shall not be included in the magnesium powder offered for acceptance.

3.1.2 Type I (grade B). Type I (grade B) magnesium powder shall be manufactured from magnesium metal of such purity that the product meets the requirements of this specification. Mica fillers or other adulterants shall not be used and "bag house dust" shall not be included in the magnesium powder offered for acceptance.

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3.2 Form.

3.2.1 Type I. Type I magnesium powder shall consist of shavings, turnings, flakes, plates or any combination of these which meet the necessary granulation requirements shown in table II.

3.2.2 Type II. Type II magnesium powder shall consist of oblong chip-like particles with rounded edges.

3.2.3 Type III. Type III magnesium powder shall consist of granular or spheroidal particles.

3.3 Chemical composition. Magnesium powder shall conform to the requirements shown in table I.

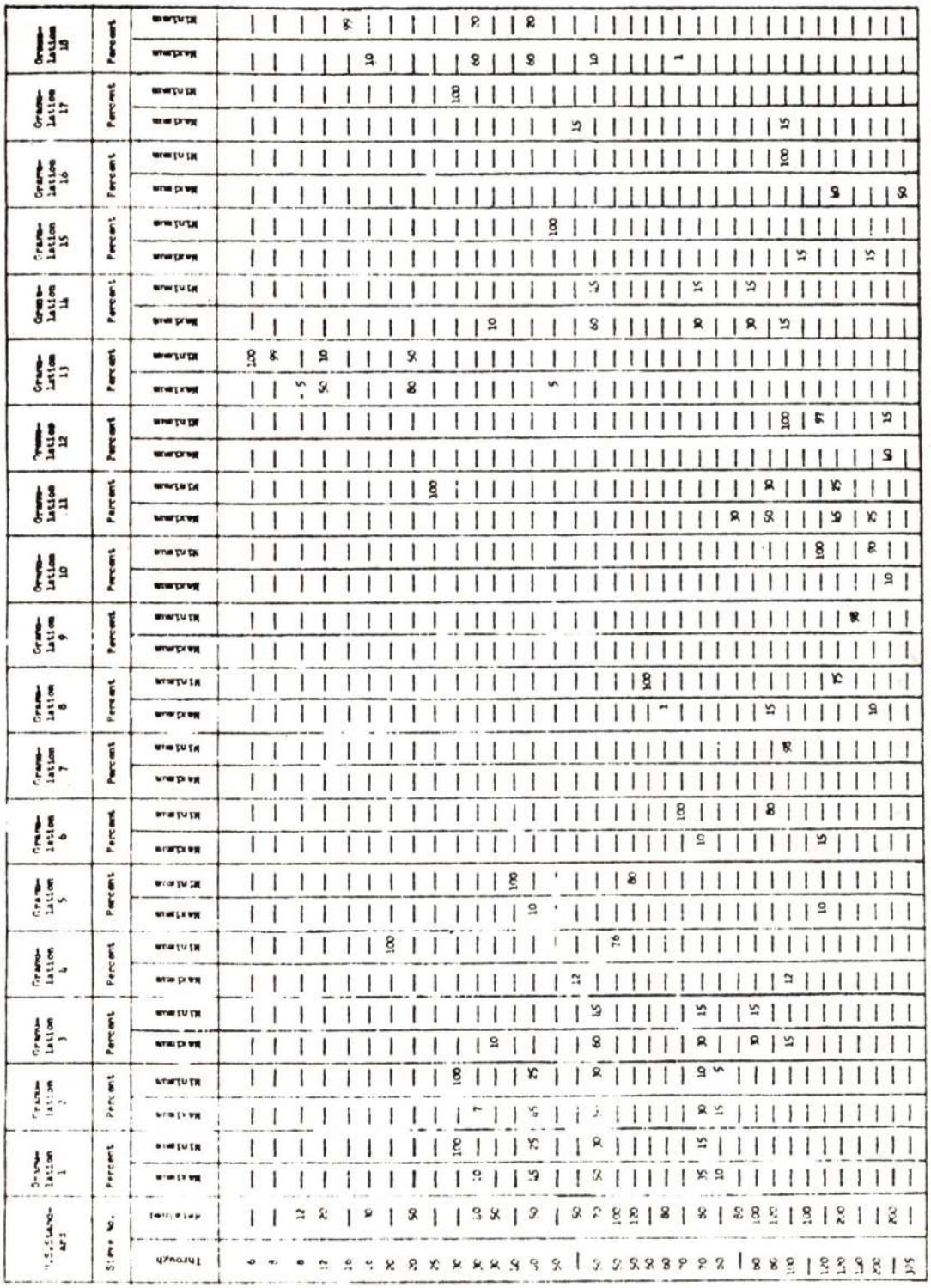
Table I. Chemical composition

	Type I		Type II	Type III
	Grade A	Grade B		
	Percent	Percent	Percent	Percent
Material volatile at 105°C (max.)	0.1	0.1	0.1	0.1
Oil and grease (max.)	.02	.02	.02	.02
Carbides (max.)	.004	.004	.004	.004
Material insoluble in sulfuric acid (max.)	.15	-	.15	.15
Metallic iron (max.)	.05	-	.05	.05
Total iron as Fe_2O_3 (max.)	.25	-	.25	.25
Aluminum as Al (max.)	-	10	-	-
Free metallic magnesium (min.)	96	-	96	98
Aluminum plus free metallic magnesium (min.)	-	96	-	-

3.4 Granulation. Magnesium powder shall be of the fineness shown in table II. All percentages shall be by weight using U. S. standard sieves conforming to RR-S-366. The powder shall pass through the required sieves readily without balling or the particles clinging together.

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Table II. Granulation



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3.5 Apparent density. The apparent density of the magnesium powder when determined as specified in 4.3.11 shall be as shown in table III.

Table III. Apparent density

Granulation No.	Density in grams per milliliter	
	Maximum	Minimum
1	0.65	0.55
2	.65	.55
3	.75	.65
4	.625	.45
5[1]	-	-
6[1]	-	-
7[1]	-	-
8[1]	-	-
9[1]	-	-
10[1]	-	-
11	-	.45
12	-	.45
13	-	.70
14	.75	.65
15	-	.90
16	-	.90
17	-	.90
18[1]	-	-

[1]No determination required.

3.6 General.

3.6.1 Analysis of metal. A chemical analysis of the metal used in the manufacture of magnesium powder shall be furnished with each lot offered for acceptance.

3.6.2 Description sheets. If the material represented in the lot comes from more than one final processing unit, the contractor shall furnish with each lot of material submitted for acceptance a description sheet giving a complete history of its manufacture. If the lot is blended before packing, this operation shall be considered as the final processing unit.

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3.7 Workmanship.

3.7.1 Process.

3.7.1.1 Types I and II. Types I and II magnesium powder shall be manufactured by grinding, stamping, turning, or any combination of these methods which will yield a suitable product.

3.7.1.2 Type III. Type III magnesium powder shall be manufactured by the atomizing process.

3.7.2 Care shall be exercised to avoid inclusion of foreign material during manufacture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the magnesium powder produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. A minimum of three containers shall be selected in such a manner as to be representative of the lot. When lots comprise less than three containers, all the containers in the lot shall be selected. If the material represented in the lot comes from more than one final processing unit, the containers selected shall fairly represent the production of each of the final processing units present in the lot. Before sampling, the contents of each selected container shall be thoroughly blended by rolling and turning end for end. By means of a thief, sufficient magnesium powder shall be removed from each selected container to form a primary sample of approximately 1 pound. This sample shall be mixed thoroughly. An 8 ounce portion of the primary sample shall be placed in a tin canister and labeled so that the container from which it was taken can be easily identified. The remaining portions of the primary sample shall be mixed thoroughly in a grounded metal container and quartered to form a composite sample of approximately 1 pound. The composite sample shall be placed in a tin canister which is nearly filled and labeled to show the name of the material, type, grade, and granulation number, manufacturer, plant, contract or order number, lot number and number of pounds in the lot. Unless otherwise specified, all acceptance tests shall be made on the composite sample representative of the lot (see 6.3). Primary samples shall be held for possible future examination.

4.2.3 Inspection methods. Unless otherwise specified, all examinations and tests shall be conducted as specified herein to determine compliance with the applicable requirements of section 3. Water in accordance with ASTM D1193 and reagent chemicals shall be used.

4.3 Tests.

4.3.1 Form. Place a small portion of the sample on a glass slide and examine under a 20 to 30 power microscope.

4.3.2 Material volatile at 105°C. Weigh into a tared dish approximately 5.000 gm. of the sample, heat for 3 hours at 105°C., cool and weigh. Calculate the percent material volatile at 105°C as follows:

$$\text{Percent material volatile at } 105^{\circ}\text{C} = \frac{100A}{W}$$

where A = gm. loss in weight
W = gm. of sample

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4.3.3 Oil and grease. Extract a 50.00 gm. sample with 250 ml. of chemically pure ethyl ether (redistilled over sodium) for 1 hour in a Soxhlet extractor or similar apparatus having an extracting tube of 250 ml. capacity. Evaporate on steam bath to a volume of approximately 50 ml. and filter the extract into a tared 150 ml. beaker. Rinse the flask with three 10 ml. portions of ethyl ether. Make a blank determination for residue on the ether. Cover the beaker with a ribbed watch glass. Evaporate the ether on a water bath that is constantly at boiling temperature. The total time of heating on the water bath should be approximately 1 hour. Remove from the water bath and dry the outside of the beaker with a clean cloth. Heat in an oven at 100° to 105°C for 30 minutes, cool and weigh. Calculate the percent oil and grease as follows:

$$\text{Percent oil and grease} = \frac{100\ (A-B)}{W}$$

where A = gm. of ether extract
B = gm. from blank
W = gm. of sample

Note. Considerable care shall be exercised through the entire procedure to prevent contamination of the sample or the extract with grease or dust from the hands or other source.

4.3.4 Carbides.

4.3.4.1 Apparatus. Use an apparatus similar to that shown in figure 1.

4.3.4.2 Hydroxylamine hydrochloride reagent. Dissolve 0.25 gm. of gelatin in hot water. Dilute to 500 ml. and add 500 ml. of 95 percent ethyl alcohol and 1.25 gm. hydroxylamine hydrochloride.

4.3.4.3 Procedure. Add a weighed portion of 50 gm. of magnesium powder to the reaction flask (A). Transfer 50 mg. of cuprous chloride to test tube (B). Pass nitrogen through the apparatus to remove the air and continue the passage of nitrogen throughout the determination. Lower the test tube and add 15 ml. of concentrated NH_4OH and 30 ml. of hydroxylamine hydrochloride reagent. Raise the test tube, insert stopper, add 100 ml. of water from dropping funnel (C) and switch on the heater. Heat the mixture in the reaction flask just to boiling. A red or pink coloration of the cuprous ammonium chloride solution in the test tube indicates that the magnesium sample contains more than 0.004 percent carbides. Should the cuprous ammonium chloride solution become blue, due to the introduction of air, it must be decolorized with successive 5 ml. portions of hydroxylamine hydrochloride reagent until the blue color disappears.

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4.3.4 Material insoluble in sulfuric acid. Transfer 5.000 gm. of the sample to a 600 ml. beaker and dissolve in 150 ml. of 1:5 H_2SO_4. Allow to stand until action ceases and all of the magnesium has dissolved. Filter through a tared crucible and wash thoroughly with small portions of hot water, being careful to keep the volume of filtrate and washings at approximately 200 ml. Reserve the filtrate for determination of total iron (see 4.3.7). Heat the crucible at 105°C for 2 hours, cool, and weigh. Calculate the percent material insoluble in sulfuric acid as follows:

$$\text{Percent material insoluble in sulfuric acid} = \frac{100A}{B}$$

where A = gm. of residue
B = gm. of sample

4.3.6 Metallic iron. Using 100 gm. of sample, pass a magnet or an electromagnet through the powder until no more particles are taken up. Brush the particles adhering to the magnet onto a sheet of glazed paper and separate the magnetic from the nonmagnetic by passing the magnet under the paper and drawing the magnetic particles aside. Repeat the procedure to eliminate as much as possible any adhering particles of magnesium. Transfer the magnetic particles onto a balance pan, weigh, and report as metallic iron. In the event that this weight exceeds the requirements for the grade of powder being tested, analyze for iron as follows: Transfer the weighed material to a 250 ml. beaker and add 100 ml. of 5 percent sulfuric acid. Boil until dissolved. Cool. Pass the solution through a Jones reductor, taking the usual precautions to activate the amalgamated zinc. Wash with 50 ml. of 5 percent sulfuric acid and 150 ml. of water. Titrate with standard 0.02N potassium permanganate to a pink color. Calculate the percent metallic iron as follows:

$$\text{Percent metallic iron} = \frac{5.58\ VN}{W}$$

where V = ml. of $KMnO_4$ solution
N = normality of $KMnO_4$ solution
W = gm. of sample

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4.3.7 Total iron. Transfer the filtrate from the determination of insoluble matter (see 4.3.5) to a 300 ml. Erlenmeyer flask, add a spiral of aluminum wire, cover with a small watch glass and boil for approximately 10 minutes. Cool under running water, remove the aluminum spiral and titrate immediately with N/20 $KMnO_4$ solution to a pink color. Run a blank determination on the aluminum using 150 ml. of 1:5 H_2SO_4. Calculate the percent total iron as follows:

$$\text{Percent total iron (as } Fe_2O_3) = \frac{7.98\ (V-B)N}{W}$$

where V = ml. of $KMnO_4$ for sample
B = ml. of $KMnO_4$ for blank
N = normality of $KMnO_4$ solution
W = gm. of sample

4.3.8 Aluminum (type I, grade B). Transfer 0.4000 gm. of the sample to a 400 ml. beaker. Cover with a watch glass and add 10 ml. of concentrated H_2SO_4 down the lip of the beaker. Add 50 ml. of water very slowly, taking care to prevent the reaction from proceeding too rapidly. When the reaction has subsided, boil to dissolve the last particles of metal. If there is any residue, add HNO_3 to clear. Filter the solution if necessary, wash and dilute to about 250 ml. Add 25 ml. of a saturated solution of NH_4CL and a few drops of methyl red indicator. Heat nearly to boiling and add 1:4 NH_4OH until the color just changes to yellow. Boil for about 1 minute and immediately filter through a Whatman No. 40 or equivalent filter paper. Wash twice with hot 2 percent NH_4Cl solution. Place the paper and precipitate back into the original beaker. Add 20 ml. of HCl (1:1), digest on steam bath for a few minutes, and break up the filter paper with a stirring rod. Dilute to about 200 ml. with water and precipitate as before. Police and wash several times with hot 2 percent NH_4Cl solution. Transfer the precipitate to a tared platinum crucible. Burn the filter paper at dull red heat and ignite at 1000° to 1100°C for 1 hour. Cool in a desiccator and weigh as rapidly as possible. Calculate the percent aluminum as follows:

$$\text{Percent aluminum} = \frac{52.93A}{W}$$

where A = gm. of residue
W = gm. of sample

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4.3.9 Free metallic magnesium.

4.3.9.1 Apparatus.

(a) 1 graduated dropping funnel.
(b) 1 generating flask.
(c) 1 20 inch reflux condenser.
(d) 1 glass-stoppered gas washing bottle containing concentrated H_2SO_4.
(e) 1 drying column containing dehydrite or equivalent.
(f) 1 combustion furnace having the tube project from the furnace approximately 6 inches on the inlet side and 2 and 1/2 to 3 inches on the outlet side. The tube shall be filled with copper oxide held in place by a small asbestos plug.
(g) 1 absorption tube containing dehydrite or equivalent.
(h) 2 guard bottles containing concentrated H_2SO_4.
(i) 1 capsule for weighing and introducing the sample. (This may be conveniently made by cutting off a small vial 3/4 inch from the bottom.)
(j) 1 suction line.

4.3.9.2 Assembly of apparatus. Connect the gas generating flask to the bottom of the dropping funnel using a Y tube having one side open as an air inlet. Connect the lower end of the reflux condenser to the generating flask and the upper end to the gas washing bottle. Connect the lower end of the drying column to the gas washing bottle and the upper end to the inlet of the combustion furnace. Connect the lower end of the absorption tube to the outlet of the combustion furnace and the upper end to the two guard bottles connected in series. Connect the outlet of the second guard bottle to the suction line.

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4.3.9.3 Procedure. Plug in the furnace, adjust the rheostats to heat the furnace quickly to 600°C. and turn on the water in the reflux condenser. In approximately 1 hour turn on the suction to draw air through the train at the rate of approximately 5 bubbles per second in the gas washing bottle. Weigh the absorption tube to obtain the initial weight, reconnect it into the train, draw air through it for 45 minutes, disconnect, cool for 30 minutes, and reweigh. A blank of 0.4 mg. or less is considered satisfactory. Transfer 1.000 gm. of the sample into the capsule and carefully slide it into the generating flask, being careful not to scatter the powder. Reconnect the absorption tube, adjust the rheostats to maintain the furnace at 580° to 620°C and carefully add approximately 60 ml. of distilled water, enough to cover the lower end of the inlet tube, through the dropping funnel. Place 30 ml. of 30 percent H_2SO_4 in the dropping funnel and begin the decomposition of the sample slowly by adding the H_2SO_4 a drop or two at a time, being careful to note that the bubbles are constantly entering the inlet tube at the rate of approximately 5 bubbles per second. It is advisable to have on hand a large beaker filled with cold water in which the generating glask may be immersed if the reaction becomes too violent. After 10 to 15 ml. of H_2SO_4 have been added, the reaction is nearly complete and the remainder of the acid may be added rapidly. Gently boil the contents of the flask for 3 minutes, draw air through for an additional 30 minutes, disconnect the absorption tube, cool, and weigh. Disconnect the generating flask and rinse the dropping funnel and the inlet tube well with distilled water to remove acid. Calculate the percent metallic magnesium as follows:

$$\text{Percent free metallic magnesium} = \frac{135\ (A-B)}{W}$$

where A = gain in weight of absorption tube
B = gm. from blank
W = gm. of sample

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4.3.9.4 Alternate procedure for determination of free metallic magnesium (eudiometer method). When desired, the following eudiometer method may be used: Assemble an apparatus similar to that shown on figure 2. The gas measuring buret shall be graduated in 0.2 ml. The reaction flask shall contain 100 ml. of 3 percent H_2SO_4 which has been saturated with hydrogen gas. Means shall be provided for dropping the sample into the acid. The weight of the sample to be used will depend on the metal content and the temperature and pressure at which hydrogen is measured. At standard conditions of temperature and pressure, a 0.47 gm. sample of pure magnesium will yield about 470 ml. of hydrogen. The weight of sample should be roughly computed to give a volume of hydrogen within the capacity of the graduated portion of the gas buret. The system must be in equilibrium with respect to temperature and pressure and checked for any gas leakage before the determination is started. The weighed sample is folded in a piece of paper and placed in position. The temperature is recorded after comparing the reading of the thermometer in the reaction flask with a thermometer located in the close vicinity of the apparatus. The volume is also recorded after the pressure is equalized by means of the leveling bulb. The confining liquid in the pipet is water saturated with hydrogen gas and containing 0.1 percent H_2SO_4 and a few drops of methyl orange to facilitate the observation of the solution in the apparatus. Allow the sample to drop into the acid and shake. After the reaction has ceased, again bring the system to equilibrium by cooling all parts of system to the initial temperature ($\pm$ 1°C) and again equalizing the pressures. Record the volume of hydrogen gas evolved. Calculate to percent free metallic magnesium as follows:

$$\text{Percent free metallic magnesium} = \frac{\frac{(P-P_W)}{760} \times \frac{273}{(273+T)} \times 0.001086 \times 100}{W} + \frac{0.03901A(P-P_W)}{W(273+T)}$$

where A = reading of gas buret in ml.
P = reading of barometer in mm. Hq.
P_W = vapor pressure of water at temperature T
T = temperature of the system at equilibrium in degrees Centigrade
W = gm. of sample

4.3.9.5 Referee method. In case of dispute, the water conversion method shall be used.

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4.3.10 Granulation. Place a weighed portion of approximately 50 gm. of the sample on the top sieve of the next of sieves assembled as specified in table II and provided with a bottom pan. Cover and shake for 30 minutes in a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the portions retained by each sieve and calculate to a percentage basis as required.

4.3.11 Apparent density. Assemble a Scott volumeter as shown on figure 3 having a No. 8 U. S. standard sieve in the upper hopper, so that the tared receiving cube, one cubic inch in volume, is directly under the lower funnel and rests on the base of the apparatus. Slowly pour approximately 15 to 20 gm. of the sample into the hopper and brush if necessary through the sieve in order to fill the receiving cube to overflowing. By means of a spatula, carefully strike off the excess magnesium from the cube, taking care not to jar the cube during this operation. Weigh the cube and contents. Three determinations shall be made and calculated. Calculate the apparent density values as follows:

$$\text{Apparent density} = \frac{W}{V}$$

where W = gm. of sample
V = volume of receiving cube in ml.

4.4 Resubmission. If the composite sample representative of the lot fails to pass the inspection tests, the manufacturer shall have the option of having an analysis of each primary sample made at no further expense to the Government. The manufacturer may then remove or replace defective portions of the lot represented by the primary samples which fail to meet the requirements, and resubmit the lot for acceptance, provided that the markings on the containers are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the inspector.

5. PREPARATION FOR DELIVERY

5.1 Application. The packing and marking requirements specified herein apply only to direct purchase by or direct shipment to the government.

5.2 Packing. Unless otherwise specified, level A packing shall be employed.

5.2.1 Level A. Unless otherwise specified in the contract or order, magnesium powder shall be packed in unlined containers conforming to MIL-C-6054, class I, IA, II or III and the requirements of Code of Federal Regulations 49 CFR 171-179.

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5.2.2 Level B. Not applicable.

5.2.3 Level C. The packing shall be adequate to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. The containers shall comply with the Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation, and the Code of Federal Regulations 49 CFR 171-179.

5.3 Marking. In addition to any other special marking required, shipments shall be marked in accordance with MIL-STD-129. Marking to insure safe handling shall conform to the Code of Federal Regulations 49 CFR 171-179.

6. NOTES

6.1 Intended use. Magnesium powder covered by this specification is intended for use in the manufacture of tracer, igniter, signal flare and pyrotechnic compositions. Type II material is required for the manufacture of bursting igniters. In general, higher density material will give longer burning times in tracers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, grade, and granulation required (see 1.2).
(c) Quantity required.
(d) The size and type of containers (see 5.).
(e) Selection of applicable packing level.

6.3 When in the opinion of the inspector, the operation of the final processing units was not uniform during the time the material in the lot was produced, he should require a sieve analysis on each primary sample (see 4.2) or on suitable sublot composite samples.

Custodian:
Army - MU

Preparing activity:
Army - MU

Review activities:
Army - MU, MD

Project No. 6810-A892

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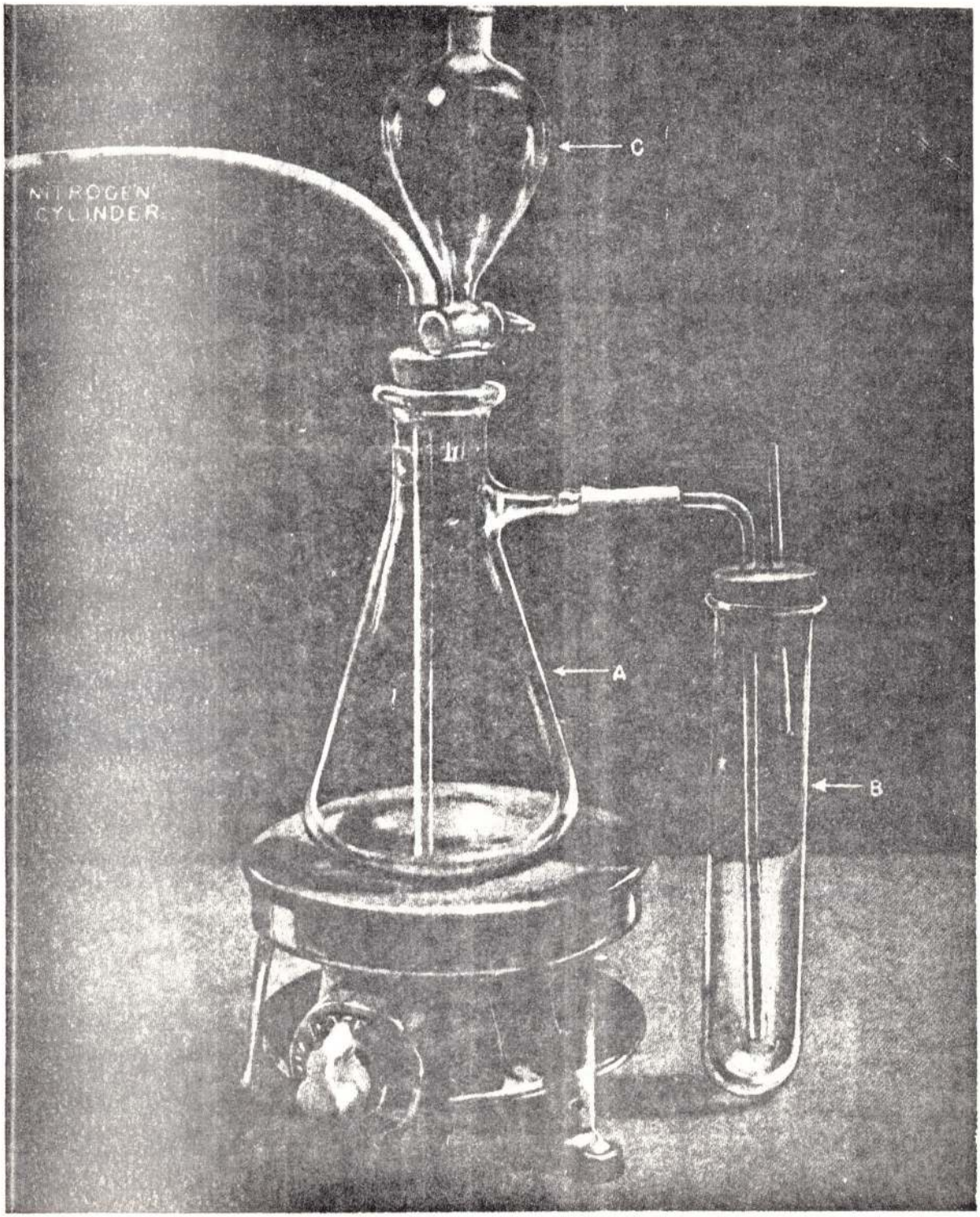


FIGURE 1.—**Apparatus for Determination of Carbides in Magnesium.**

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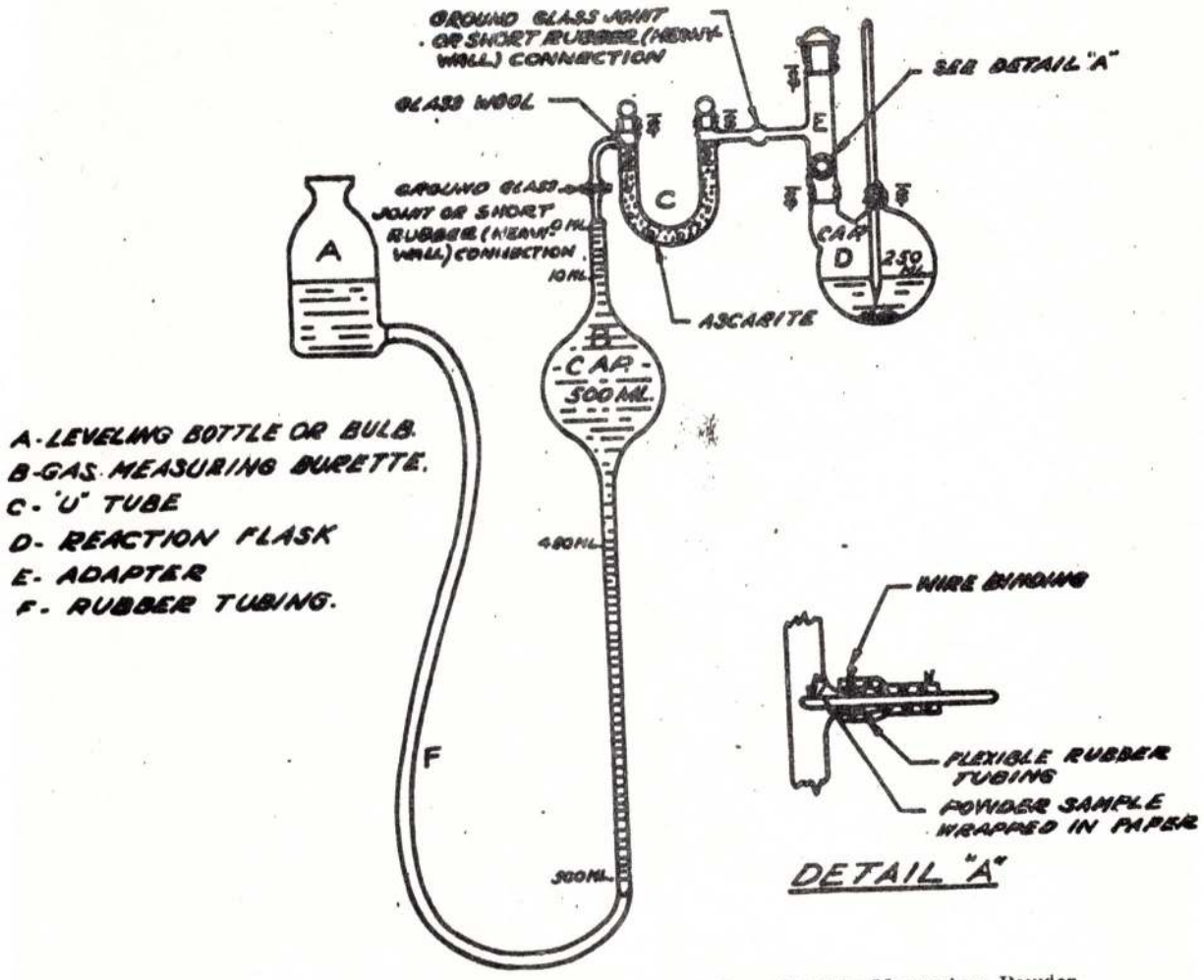


FIGURE 2.—Eudiometer for Determination of Magnesium Metal in Magnesium Powder.

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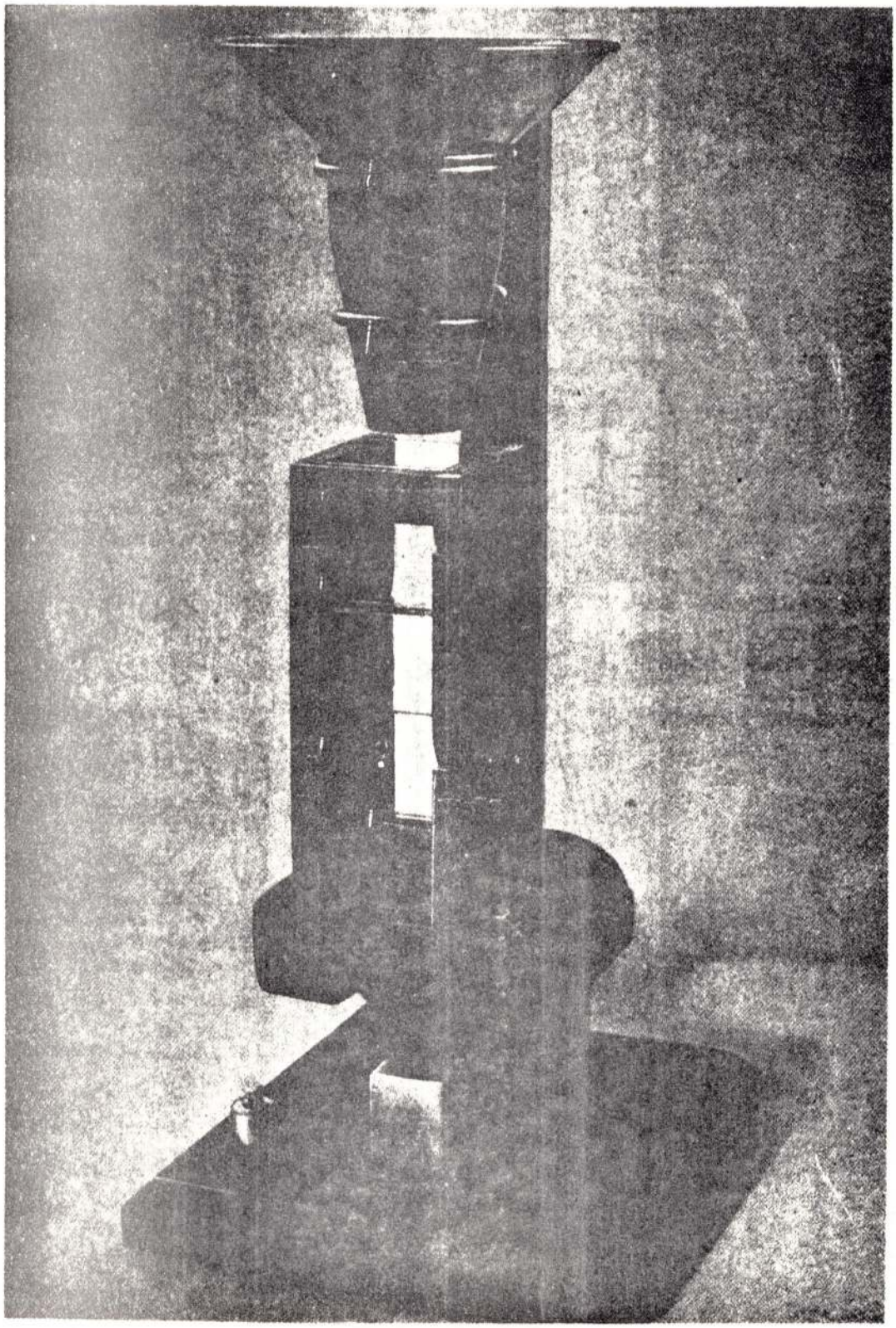


FIGURE 8.—Apparatus for Determination of Apparent Density.

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